Obstructions and a Corrected Spectral Criterion for GRH

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Research draft. GRH is not proved here.

Abstract

We audit proposed surface-order and Hilbert–Pólya arguments for GRH. We prove obstructions to the proposed prime/geodesic identification, integral cyclic induction, absolute prime-tail estimates and determinant preservation, and retain a conditional spectral closure criterion. We construct convergent arithmetic approximants with explicit errors and analyze coefficient-defined Jensen metrics. From the explicit scalar local factors of a separate exponent argument, we extend an Euler-correction region and its principal-residue interface, and isolate a shifted infinite-order angular Hecke factor. Under the stated classical analytic inputs, its zeros obstruct extending the unchanged correction's nonvanishing criterion below 7/12. We also reconstruct a frozen-capacity row count conditionally on the source moment and witness estimates, and extend the low interface under explicit reflected-energy and Gram hypotheses. The nonprincipal central contours also extend under their stated arithmetic and buffered-bound contracts. We verify the finite-field Gauss and zero-preserving correlation input, with prime-power and reciprocity dependencies distinguished. We also prove an annular periodic cancellation lemma and reconstruct the Gram summation under explicit arithmetic representation and period inputs. A classical conductor argument supplies an explicit common arithmetic period, distinguished from the smaller quadratic-phase modulus. We verify the local reflected coefficient separation and its common-profile normalization, retaining the global theta and hybrid-sieve dependencies. A pinned Lean 4 companion verifies selected arithmetic, complex-analytic, matrix and infinite-product components, with its remaining dependencies specified. These results do not establish GRH; the full arithmetic spectral bridge and the required mean-square and contour estimates remain unresolved.

1 Scope and conventions

For a primitive Dirichlet character χ of conductor q, write

$$L(s,\chi)=\sum_{n\ge 1}\frac{\chi(n)}{n^s},\qquad \Lambda(s,\chi)=\left(\frac{q}{\pi}\right)^{(s+a)/2}\Gamma\left(\frac{s+a}{2}\right)L(s,\chi),\quad a=\frac{1-\chi(-1)}{2}.$$

The series and Euler product initially apply for $\operatorname{Re} s>1$. Our target is Dirichlet GRH: every zero in $0<\operatorname{Re} s<1$ has real part $1/2$. For $q=1$, use the entire function $\xi(s)=\frac{1}{2}s(s-1)\Lambda(s,1)$ when an entire target is required. Finite-order Hecke characters and general Artin representations require separate constructions; a theorem for one family is not silently a theorem for another.

The supplied surface-order manuscripts [1, 2] propose arithmetic transfer operators and self-adjoint approximants. We isolate valid closure logic and prove obstructions to particular claimed constructions. These obstructions refute intermediate statements, not GRH. The separate `math`

repository [3] includes claimed zero-free regions strictly weaker than GRH. Such a region cannot replace the missing critical-line argument.

2 A corrected conditional closure theorem

Theorem 2.1 (Spectral approximation criterion)**.** *Let $U \subset \mathbb{C}$ be connected and open, let f be holomorphic on U and not identically zero, and let*

$$F_N(s) = e^{a_N + b_N s} \det\bigl(sI - (\tfrac{1}{2}I + iH_N)\bigr), \qquad H_N = H_N^* \in M_{d_N}(\mathbb{C}).$$

If $F_N \to f$ locally uniformly on U, then every zero of f in U has real part $1/2$.

Proof. The eigenvalues $\lambda_{j,N}$ of H_N are real. Thus the zeros of F_N are $1/2 + i\lambda_{j,N}$, with algebraic multiplicities; the exponential factor never vanishes. If $f(z) = 0$ off that line, choose a closed disk $D \subset U$ about z disjoint from the line, with f nonzero on its boundary. Such a disk exists because the zeros of a nonzero holomorphic function on a connected domain are isolated. Let $m = \min_{\partial D} |f| > 0$. Local uniform convergence gives $|F_N - f| < m$ on ∂D for sufficiently large N. Rouché's theorem forces F_N and f to have the same positive number of zeros inside D, contradicting the zero location of F_N. □

Remark 2.2. This is a conditional criterion, not a construction. Self-adjointness alone establishes only the zero location of the approximants. The difficult arithmetic statement is convergence to the particular completed L-function. Pointwise convergence, weak operator convergence, and agreement of finitely many zeros do not supply the displayed hypothesis.

Proposition 2.3 (A usable continuation interface)**.** *Suppose U is a connected domain, the entire functions F_N are locally uniformly bounded on U, and they converge pointwise on a nonempty open subset $V \subset U$ to a holomorphic function f defined on U. Then $F_N \to f$ locally uniformly on U.*

Proof. Montel's theorem makes the family normal. Every subsequence has a further subsequence converging locally uniformly to a holomorphic function g. On V, pointwise convergence gives $g = f$, so the identity theorem gives $g = f$ on U. If the original sequence failed to converge locally uniformly, a compact set, positive error, and violating subsequence would contradict this subsequence property. □

A proposed analytic continuation proof must actually provide the uniform bounds and a genuine overlap V. The disjoint domains $\operatorname{Re} s > 1$ and $0 < \operatorname{Re} s < 1$ are not an overlap.

3 Arithmetic obstructions to the claimed completion

Theorem 3.1 (No integral prime-power geodesic norms)**.** *Let $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ be hyperbolic. Its expanding eigenvalue in absolute value is $u > 1$, and its hyperbolic norm is $N(\gamma) = u^2$. Then $N(\gamma)$ is not an integer greater than one. In particular no closed geodesic arising from such a matrix has length $k \log p$ for a prime p and integer $k \geq 1$.*

Proof. Put $t = |\operatorname{tr} \gamma| \in \mathbb{Z}$, so $t = u + u^{-1}$. If $u^2 = n \in \mathbb{Z}$, $n > 1$, squaring gives

$$t^2 = n + 2 + \frac{1}{n}.$$

The left-hand side and $n + 2$ are integers, so $1/n$ is an integer, contradicting $0 < 1/n < 1$. Since length is $\log N(\gamma)$, length $k \log p$ would imply $N(\gamma) = p^k$, already excluded. □

The statement applies to every subgroup of $\mathrm{SL}_2(\mathbb{Z})$, hence to congruence surfaces. The prime geodesic theorem is an asymptotic counting analogy; it cannot provide the termwise identification claimed in the completion section of [1]. Hecke correspondences of determinant p must not be identified with hyperbolic deck transformations of determinant one.

Theorem 3.2 (Failure of integral cyclic induction)**.** *The trivial character of $G = C_2 \times C_2$ is not an integer linear combination of characters induced from one-dimensional characters of cyclic subgroups of G.*

Proof. Every cyclic subgroup has order one or two. The degree of any such induced character is its index, namely four or two. Evaluation at the identity sends every integer linear combination to an even integer, whereas the trivial character has degree one. □

This distinguishes rational Artin induction from integral Brauer induction. It invalidates the stated cyclic integral reduction, while leaving appropriate forms of these classical induction theorems available for a revised argument.

Theorem 3.3 (Prime-tail obstruction)**.** *For every real $\sigma \le 1$, the positive series $\sum_p p^{-\sigma}$ diverges. Thus $\sum_{p>N} p^{-\sigma}$ is not a vanishing absolutely convergent tail in the critical strip.*

Proof. It suffices to show $\sum_p 1/p = \infty$, since $p^{-\sigma} \ge p^{-1}$. Suppose the reciprocal-prime series converged. For $0 \le x \le 1/2$, $-\log(1-x) \le 2x$, for example by integrating $(1-t)^{-1} \le 2$ from zero to x. Hence the finite Euler products $\prod_{p\le X}(1-1/p)^{-1}$ would be bounded uniformly in X. Expanding each as a finite product of absolutely convergent geometric series gives the sum of $1/n$ over all positive integers with prime factors at most X. In particular it dominates $\sum_{n\le \lfloor X \rfloor} 1/n$, which is unbounded. Contradiction. □

Any repair below $\operatorname{Re} s = 1$ must introduce proved cancellation or a regularization, with quantitative control; absolute prime-tail bounds cannot suffice.

4 Determinants and lost arithmetic information

For $T = \operatorname{diag}(\chi(p)p^{-s})_{p\le N}$ one has exactly

$$\det(I - T) = \prod_{p\le N} (1 - \chi(p)p^{-s}) = L_N(s,\chi)^{-1},$$

not L_N. On an absolute-convergence region,

$$\log\det(I - T) = -\sum_{k\ge 1} \frac{\operatorname{tr}(T^k)}{k}.$$

The reciprocal orientation matters before any analytic continuation. Moreover Hermitian-part extraction does not preserve this determinant:

$$T = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \frac{T + T^*}{2} = 0, \quad \det(I - T) = 2 \ne 1 = \det(I).$$

An identity for the transfer determinant therefore does not automatically identify the centered characteristic polynomial of a symmetrized operator.

Proposition 4.1 (A direct obstruction to the supplied diagonal diagnostic)**.** *For the $q = 1$ model evaluated at $s = 1/2$, the algorithm that sets $\lambda_p = -\arg(p^{-1/2})$ gives $H_N = 0$ and $F_N(s) = (s - 1/2)^{d_N}$. These functions cannot converge locally uniformly to $\Lambda(s, 1)$ or $\xi(s)$ on a neighborhood of $1/2$.*

Proof. Every $p^{-1/2}$ is positive, so its argument is zero and $F_N(1/2) = 0$. On the other hand the alternating Dirichlet series $\eta(1/2) = \sum_{n\geq 1}(-1)^{n-1}n^{-1/2}$ is positive: grouping into adjacent pairs gives positive terms and its sum is at least $1 - 1/\sqrt{2} > 0$. The continuation identity $\eta(s) = (1 - 2^{1-s})\zeta(s)$ gives $\zeta(1/2) < 0$. The gamma factor at $1/4$ is positive and finite, so $\Lambda(1/2, 1) \neq 0$, and its pole-removal factor at $1/2$ is nonzero. Even pointwise convergence at that point is impossible. □

Proposition 4.2 (Holomorphic Hermitian rigidity)**.** *Let $U \subset \mathbb{C}$ be connected and open, with a fixed inner product on $\mathbb{C}^d$. If $H : U \to M_d(\mathbb{C})$ has holomorphic entries and $H(s)^* = H(s)$ for every complex $s \in U$, then H is constant.*

Proof. In fixed orthonormal coordinates, for each vector v, the holomorphic scalar $v^*H(s)v$ is real-valued, so it is constant by the open mapping theorem. Polarization recovers all matrix entries from these quadratic forms, hence every entry is constant. Equivalently, $H_{ij} + H_{ji}$ and $i(H_{ij} - H_{ji})$ are real-valued holomorphic functions whose constancy determines H_{ij}. □

Thus a nonconstant arithmetic transfer $T(s)$ cannot retain entrywise holomorphy after pointwise Hermitian-part extraction: a holomorphic $(T + T^*)/2$ would be constant, and the remaining holomorphic skew-Hermitian part would be constant as well. This excludes that particular shortcut, not fixed self-adjoint operators. The spectral criterion uses H_N independent of the spectral variable s. Requiring self-adjointness only on a real parameter line is also consistent with a nonconstant holomorphic extension, but needs its own zero-location theorem.

Proposition 4.3 (A positive generalized spectral pencil)**.** *Let $A = A^*$ and $B = B^* > 0$ be fixed $d \times d$ matrices. Every zero of $\det(A - zB)$ is real. Consequently every zero of $\det(A + i(s - 1/2)B)$ lies on $\operatorname{Re} s = 1/2$.*

Proof. Singularity supplies $v \neq 0$ with $Av = zBv$. Hence $z = (v^*Av)/(v^*Bv)$, with real numerator and strictly positive real denominator. Thus z is real. Substitute $z = -i(s - 1/2)$. □

This supplies a valid finite spectral mechanism, but not its arithmetic construction. To apply it one must construct A_N, B_N, certify $B_N > 0$, and prove the required completed-L determinant approximation independently. Hermitian coefficients alone are insufficient: for

$$A = \begin{pmatrix} 0 & \varepsilon \\ \varepsilon & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \det(A - zB) = -(z^2 + \varepsilon^2),$$

$0 < \varepsilon < 1/2$ gives transported off-line zeros $s = 1/2 \pm \varepsilon$ inside the critical strip.

5 A necessary change of approximation class

Theorem 5.1 (Zero-free Euler truncations cannot approximate a zero)**.** *Let U be a connected open subset of $\{\operatorname{Re} s > 0\}$, let g_N be holomorphic and nowhere zero on U, and let P_N be arbitrary finite sets of rational primes. Put*

$$D_N(s) = \prod_{p \in P_N} (1 - \chi(p)e^{-s \log p}).$$

Each of $g_N D_N$ and g_N/D_N is holomorphic and nowhere zero on U. If either sequence converges locally uniformly to a holomorphic f which is not identically zero, then f has no zero in U.

Proof. For $\operatorname{Re} s > 0$, $|\chi(p)e^{-s\log p}| \le p^{-\operatorname{Re} s} < 1$, so every local factor is nonzero. Hence the finite product, its reciprocal, and their normalized versions have the asserted property. If f had a zero, choose a closed disk inside U with a positive number of zeros of f in its interior and none on its boundary. Local uniform convergence and Rouché's theorem would force the zero-free approximants to have that positive number of zeros, a contradiction. □

More quantitatively, on any such disk D, every zero-free holomorphic approximant G satisfies

$$\max_{\partial D} |G - f| \;\ge\; \min_{\partial D} |f| > 0.$$

Otherwise Rouché yields the same contradiction. Thus improved estimates cannot repair this approximation class near a nontrivial zero. The construction must change so that finite approximants can have zeros in the strip; a nonvanishing gamma or exponential normalization cannot do this.

6 A convergent arithmetic replacement

For a nonprincipal character modulo q, put $A(x) = \sum_{n\le x} \chi(n)$ and $B = \sup_x |A(x)| \le q$. Complete periods sum to zero, so B is finite. Partial summation gives, for $\sigma = \operatorname{Re} s > 0$,

$$L(s,\chi) - \sum_{n\le N} \chi(n) n^{-s} = -A(N)N^{-s} + s\int_N^\infty A(x)x^{-s-1}\,dx,$$

and therefore

$$\left| L(s,\chi) - \sum_{n\le N} \chi(n) n^{-s} \right| \le B\left(1 + \frac{|s|}{\sigma}\right) N^{-\sigma}.$$

The endpoint tends to zero and the integral is absolutely convergent. The limit agrees with the Euler definition on $\operatorname{Re} s > 1$, hence with its analytic continuation. Multiplication by the correct gamma/conductor factor gives locally uniform approximation on $\operatorname{Re} s > 0$ for either parity.

For zeta, reproduce the pole before removing it. Set

$$Z_N(s) = \sum_{n\le N} n^{-s} + \frac{N^{1-s}}{s-1}.$$

Euler summation gives, for $\sigma > 0$ and $s \ne 1$,

$$\zeta(s) - Z_N(s) = -s\int_N^\infty \{x\}x^{-s-1}\,dx, \qquad |\zeta(s) - Z_N(s)| \le \frac{|s|}{\sigma} N^{-\sigma}.$$

The integral continues holomorphically to $\sigma > 0$. Consequently

$$\xi_N(s) = \frac{s}{2}\pi^{-s/2}\Gamma(s/2)\left((s-1)\sum_{n\le N} n^{-s} + N^{1-s}\right)$$

is holomorphic on that half-plane, including $s = 1$, and converges locally uniformly to ξ; in fact $\xi_N(1) = 1/2$ for every N. These constructions use known arithmetic coefficients, not unknown zeros. They avoid Theorem 5.1 by allowing additive cancellation and unrestricted finite-level zeros. They establish convergence, but provide no self-adjoint realization or critical-line theorem for those zeros.

7 A certified test of the arithmetic pencil candidate

Let χ be the primitive nonprincipal character modulo 3. Its fifth Dirichlet partial sum is the entire function

$$f(s) = S_5(s,\chi) = 1 - 2^{-s} + 4^{-s} - 5^{-s}.$$

Theorem 7.1 (Certified zero of a finite partial sum)**.** *Put*

$$c = (337261 + 11027504i)/10^6, \qquad r = 10^{-4}.$$

The disk $|s-c| < r$ contains exactly one zero of f, counted with multiplicity, and lies inside $0 < \operatorname{Re} s < 1/2$.

Proof. The exact rational procedure detailed in the extended edition establishes

$$|f(c)| < 10^{-6}, \qquad |f'(c)| > 1, \qquad \sup_{|s-c|\le r} |f''(s)| \le 9.$$

It uses 60 terms of the logarithm's positive atanh series and terms 0 through 120 of the exponential Taylor series, with explicit remainders; no floating-point value is used in these inequalities. On the boundary, Taylor's formula gives

$$|f(s) - f'(c)(s-c)| < 10^{-6} + \tfrac{9}{2}r^2 < r < |f'(c)(s-c)|.$$

Rouché gives exactly one zero, which is therefore simple. The real part throughout the closed disk belongs to $[337161, 337361]/10^6 \subset (0, 1/2)$. □

The completed fifth partial sum has the same zero because its conductor/gamma factor is holomorphic and nonzero on this disk. It cannot equal a nowhere-zero holomorphic factor times the determinant of a fixed positive Hermitian pencil there. This rejects that exact finite-level identification. It does not reject convergence of the partial sums, modified approximants, later truncations, subsequences, or GRH itself: f is a finite partial sum, not the full L-function.

8 Jensen approximants and a coefficient-defined metric

This alternative currently concerns the Riemann target only. Define

$$\gamma_j = \frac{j!\,\xi^{(2j)}(1/2)}{(2j)!}, \qquad F(w) = \sum_{j\ge 0} \frac{\gamma_j}{j!} w^j, \qquad \Xi(t) = \xi(1/2 + it) = F(-t^2).$$

The classical positive Fourier kernel of Ξ gives $\gamma_j > 0$ [9]; evenness and entirety of ξ make F entire. Set, for $d \ge 1$,

$$J_d(x) = \sum_{j=0}^{d} \binom{d}{j} \gamma_j x^j, \qquad Q_d(w) = J_d(w/d), \qquad P_d(t) = Q_d(-t^2).$$

Proposition 8.1 (Explicit convergence without a zero-location premise)**.** *For $R, T \ge 0$,*

$$\sup_{|w|\le R} |Q_d(w) - F(w)| \le \frac{R^2 F''(R)}{2d}, \qquad \sup_{|t|\le T} |P_d(t) - \Xi(t)| \le \frac{T^4 F''(T^2)}{2d}.$$

Proof. The coefficient multiplier $w_{d,j} = j!\binom{d}{j}/d^j$ for $j \le d$, and 0 for $j > d$, satisfies $0 \le w_{d,j} \le 1$ and $1 - w_{d,j} \le j(j-1)/(2d)$. For $j \le d$ use $1 - \prod_{k<j}(1-k/d) \le \sum_{k<j} k/d$; for $j > d$ use $j(j-1) \ge 2d$. Summing the absolutely convergent series with nonnegative coefficients proves the bounds. □

Let C_d be the multiplication-by-x companion of the monic J_d/γ_d, and $H_d = (m_{i+j})_{0\le i,j<d}$, where $m_k = \operatorname{tr}(C_d^k)$ and $m_0 = d$. Newton identities compute H_d from the coefficients, and $H_d C_d = C_d^T H_d$. The Hermite–Sylvester criterion gives $H_d \ge 0$ exactly when J_d is real-rooted [8]; then its positive coefficients force all roots negative. If $H_d > 0$, the extended edition constructs an explicit real symmetric K_d with

$$P_d(t) = \frac{(-1)^d \gamma_d}{d^d} \det(tI - K_d).$$

Strict positivity is needed for that construction; semidefinite metrics can be singular. The classical Jensen criterion makes $H_d \ge 0$ for every d equivalent to zeta RH [7]. This identifies the missing positivity; it does not prove it. Fixed-degree eventual hyperbolicity as the shift grows does not settle this unshifted growing-degree family.

Proposition 8.2 (Positive smooth kernels do not supply the metric)**.** *Positivity of an even smooth Fourier density, including Gaussian decay, does not imply positivity of its Jensen Newton-sum metric.*

Proof. Convolve $\frac{9}{10}\delta_0 + \frac{1}{20}(\delta_{-1} + \delta_1)$ with a centered Gaussian of variance $v > 0$. Its Fourier transform is $e^{-vt^2/2}(9/10 + \cos t/10)$ and its density is strictly positive, even and Schwartz. Writing its associated $F(w) = \sum \gamma_j w^j/j!$ gives $\gamma_0 = 1$, $\gamma_1 = (v + 1/10)/2$ and $\gamma_2 = (3v^2 + 3v/5 + 1/10)/12$. Thus $\gamma_1^2 - \gamma_0\gamma_2 = -7/1200 < 0$; the degree-two Newton-sum metric has negative determinant. These are model coefficients, not the arithmetic ξ coefficients. □

9 Extending an Euler correction and locating an additional factor

We now test one missing analytic input of the September argument in [3]. All deductions in this section use its displayed scalar local identity; its preceding Gauss/reciprocity derivation and main mean-square theorem are not certified here.

Proposition 9.1 (Extension of the explicit local product)**.** *The displayed correction product $\mathcal{H}_{\eta,u}(x,w,z)$ converges normally on a neighborhood of*

$$\operatorname{Re} x \ge 3/4, \qquad \operatorname{Re} w \ge 19/20, \qquad \operatorname{Re} z \ge 33/200,$$

and is $O_\epsilon(q_u^\epsilon)$ there. On the displayed closed region, for $Q = Np$, its local defects are at most $1024Q^{-149/100}$ when $p \nmid u$ and $32Q^{-7/10}$ when $p \mid u$. The latter is an upper bound, not a nonvanishing statement.

Proof. Substitution in the explicit local formulas gives the exponent and denominator bounds detailed in the extended edition. All five positive row valuations are included. The ideal count sums the first bound; the divisor-product estimate handles the finitely many primes of u. The bounds retain a positive margin on a larger open region. □

For the principal row put $H_\eta(s) = \mathcal{H}_{\eta,1}(s, 1, 1/6)$. At a prime set

$$v = Q^{-1}, \quad d = \eta(p)Q^{-s}, \quad \theta(p) = \alpha(p)^{-6}\eta(p)^6, \quad r = \theta(p)Q^{3-6s},$$

where $\alpha(p) = p/|p|$ for a generator. Sixth powers remove unit ambiguity. Exact simplification gives

$$H_p = \frac{(1-v^2)[1-vr-(1-v)d]}{(1-r)(1-d)}, \qquad A_p = (1-r)H_p.$$

For $\operatorname{Re} s > 1/2$, the numerator is nonzero because $|vr + (1-v)d| < v + (1-v) = 1$. Moreover

$$A_p - 1 = -v^2 + \frac{(1-v^2)v(d-r)}{1-d}.$$

Thus $A = \prod_{p \notin S} A_p$ is holomorphic and nonzero on $\operatorname{Re} s > 1/2$, while H_η itself converges normally and is nonzero on $\operatorname{Re} s > 2/3$. On the latter region the exact factorization is

$$H_\eta(s) = A(s) L_F^S(6s-3, \theta).$$

The character θ has nontrivial angular factor α^{-6}; it has infinite order and is outside the source's finite-order target family. The correction can be made uniformly close to 1 on each fixed $\operatorname{Re} s \geq \sigma_0 > 2/3$ by a sufficiently large finite deletion. This supplies the proposed tiny perturbation's local Euler input; the compensated contour errors and mean squares remain unproved.

Proposition 9.2 (Limitation of the unchanged correction)**.** *Assume the scalar identity and the classical Hecke completion properties listed in the extended edition, including order one and conjugation. The analytic continuation of this unchanged H_η has a zero with real part at least $7/12$. It therefore cannot be nonzero throughout $\operatorname{Re} s > \sigma_0$ for any $\sigma_0 < 7/12$.*

Proof. The entire order-one completion of $L(s, \theta)$ has infinitely many zeros: otherwise Hadamard factorization would contradict its $\sigma \log \sigma + O(\sigma)$ logarithmic growth on the positive axis. Euler nonvanishing, the functional equation and conjugation give a zero ρ with $1/2 \leq \operatorname{Re} \rho \leq 1$. Neither gamma nor deleted Euler factors vanish there. Hence the displayed continuation vanishes at $(\rho+3)/6$, where A is nonzero and $\operatorname{Re} s \geq 7/12$. □

The Hecke analytic inputs are classical external dependencies, not Lean theorems here [10, 11]. No GRH or Artin holomorphy premise is used. This limits the specified probe and nonvanishing criterion; it does not refute GRH, the claimed 7/8 result, or different constructions.

10 The compensated principal residue at perturbed parameters

We retain the source's scalar local identity, exact high representation, Mellin integrated/trace tail contracts, and polynomial reciprocal control on $\operatorname{Re} s = \beta_* + e$. The following verifies a principal interface, not the remaining energy or row-count estimates. On the enlarged Euler region, the four local multiplier errors have powers

$$Q^{-x_r}, \quad Q^{-6z_r}, \quad Q^{4-5x_r-6z_r}, \quad Q^{1-w_r-6z_r}.$$

Their common saving is at least $37/50$. The local lower bound for H_p therefore gives $\mathcal{B}_p = -1 + O(Q^{-37/50})$, uniformly in all heights. On fixed real boxes the full principal correction is $O(Z^{\ell \operatorname{Re} z})$, with height degree zero.

For a nonempty fixed slot system with positive lengths ℓ_i, set

$$S_i = \sum_{p \in \mathcal{P}_i(Z)} W_i(Q/P_i) Q^{-5/6}, \qquad A_T = (-1)^K Z^{-\ell/6} \prod_i S_i, \qquad \mu = \tfrac{7}{10} \min_i \ell_i > 0.$$

Fixed-ray prime counting gives $S_i > 0$ eventually and $A_T \sim c_T(\log Z)^{-K}$ with $c_T \neq 0$, of sign $(-1)^K$ [12]. No uniformity in a moving ray conductor is used. Nonnegative weights give the exact residue factorization

$$\mathfrak{H}_{\eta,1,Z}(s,1,1/6) = H_\eta(s)Z^{\ell/6}A_T(Z)(1+\mathcal{R}_{\eta,Z}(s)), \qquad \mathcal{R}_{\eta,Z} = O(Z^{-\mu}).$$

Indeed the ith relative error is the negative weighted average of $\mathcal{B}_p + 1$, of size $O(P_i^{-37/50})$; a fixed finite product preserves that saving. Also $|A_T|^{-1} \ll_\epsilon Z^\epsilon$.

With the explicitly inherited contracts above, the source's principal contour comparison now extends to $3/4 \le \sigma_0 < \beta_*$. Its three remainders, relative to $C(\beta_*)$, have exponents

$$(1+h)e - l_y/20 + \epsilon, \qquad e - h/600 + \epsilon, \qquad e - \mu + \epsilon.$$

The extended edition derives these by crossing only $w = 1$ and $z = 1/6$. All three are negative for sufficiently small pretarget e, ϵ. The perturbed geometry at $t = 10^{-5}$ satisfies this interface, with $C(s) = s - 11/16$ and target $349999/400000$. This is not a new zero-free theorem: other low and nonprincipal contour inputs remain unverified; the following counting interface is conditional on its moment inputs.

There is a separate pointwise limitation. At the principal residue, putting $N = 1 - vr - (1-v)d$ gives

$$\mathcal{B}_p + 1 = \frac{(1-v)[r/d - vr - (1-v)d]}{N}.$$

For fixed $1/2 < \sigma = \operatorname{Re} s \le 3/5$ its norm is asymptotic to $Q^{3-5\sigma}$, uniformly in phases. In particular, for $\eta = 1$ and an inert rational prime, $\alpha^{-6} = 1$ and real $s = 3/5$ give $r = d$, so $\mathcal{B}_p = -1/Q$ exactly. The error around -1 tends to 1. Such primes can occur in the fixed identity ray. This excludes the pointwise power-saving argument there, without excluding cancellation after prime averaging or a different normalization.

11 A frozen-capacity counting interface

The next proposition concerns the September probe, not the spectral construction. Its arithmetic moment and witness inputs remain uncertified. The previously claimed 7/8 bound is an explicit hypothesis here.

Proposition 11.1 (Conditional row count)**.** *Assume $\beta_* \le 7/8$ and the source's marked inverse moment, plain fourth moment, sixth-power amplification and simultaneous saturated witness statements, with their common profiles, masks and height bounds. For each $\epsilon > 0$, first choose capacity decrements and a slot-mesh threshold independent of the eventual fixed slot count. Then fix a finite slot system satisfying it and a retained nonfloor bin, $a > 51/100$, of sixth-power-free rows of norm $\asymp U = Z^d$, where $d \ge d_{\min} > 0$. Put $\delta = 2a - 1 \le 3/4$, $x = q/\delta \in [0, 1/2]$, where q is its length-weighted prime amplitude. Suppose the total available prime-slot length ℓ/d exceeds 7/37 by a fixed margin. Then*

$$\#\mathcal{B} \ll U^{R_*+\epsilon}(1+T_1)^A, \quad R_* = 1 - \delta + \frac{(\alpha-\delta)\delta P}{2J}, \quad \alpha = 5/6,$$

where $D = 3 - 17x/9$, $P = (2 - 8x/9)(1-x)$ and $J = (\alpha-\delta)D + \delta P > 0$. The finite height order A is fixed before the external tail order. Without selected prime slots one instead has exponent $1 - 2\delta/3$, with no supply assumption.

Proof. The plain moment permits $\kappa = 3/4$ since $\beta_* \le (1+\kappa)/2$. Its exact zero-loss capacity is $z_P(m) = 2(1-2m)/9$; the inverse capacity is $z_M(r) = (1-r)/2$. Thus the source's $\Delta/4$ comparison loss is absent, rather than continued to a negative bootstrap parameter. At detector cutoff $\tau \in [1, 3/2]$, the saturated witnesses have $r+m \ge \tau - o(1)$, $|M_r|^2 \gg U^{\delta r-\epsilon}$ and $|S_m|^2 \gg U^{\delta m-\epsilon}$. Selecting whole positive slots of requested length z gives squared gain $U^{2qz-\epsilon_{\mathrm{round}}}$, with the fixed rounding loss $\epsilon_{\mathrm{round}}$ allocated within ϵ. Division of the moments by these spikes, using $|S_m|^4$ for the plain moment, gives the affine short counts

$$A_I(r) = 1 - \delta[x + (1-x)r], \quad S_\tau(r) = 1 - \delta[4x/9 + (2 - 8x/9)(\tau - r)].$$

They cross at $r_* = [(2 - 8x/9)\tau - 5x/9]/D$, with $23/37 \le r_* \le 1$ and $1/3 \le \tau - r_* \le 1/2$. Choose the plain count below r_* and the inverse count above it. Their common upper bound is $R_{\mathrm{short}}(\tau) = 1 - \delta + (\delta P/D)(3/2 - \tau)$. The requested inverse length is at most $7/37$, and the plain length at most $2/27 + o(1)$. A fixed capacity decrement preserves the strict marked-moment margins. Near zero capacity use the no-slot moments. For $r \ge 1$, amplification gives $L(\tau) = 1 - \delta + (\alpha - \delta)(\tau - 1)$. The maximum of these short and long bounds is minimized at $\tau = 1 + \delta P/(2J)$, giving R_*. With no selection, use $\tau = 1$; the short lines cross at $r = 2/3$, giving $1 - 2\delta/3$. All rounding and witness losses fit the prescribed ϵ. □

For $\ell_t = 1/6 + t$, $h_t = 13/16 + 3t/2$, the supply holds uniformly through $d \le h_t + \zeta$ whenever

$$0 \le \zeta < \frac{37\ell_t}{7} - h_t = \frac{23}{336} + \frac{53t}{14}.$$

This closes the frozen-capacity counting interface at $t = 10^{-5}$ under its explicit inputs. It does not certify those arithmetic moments, the earlier 7/8 theorem, the low reflected-energy application or the remaining nonprincipal contours. Floor-bin rows need separate estimates.

12 The perturbed low estimate

Proposition 12.1 (Conditional low interface)**.** *Assume the source's general reflected-energy estimate, completed-row and sector identities, quantitative Gram bound and exact finite low separation, with their stated coefficient, support, smoothness and height uniformities. For $0 \le t \le 1/30$, put*

$$\ell_t = 1/6 + t, \quad M_t = 5/6 - t, \quad l_x = 17/48 - t/2, \quad l_y = 23/48 - t/2.$$

For every fixed finite system of disjoint physical prime slots with positive lengths of total ℓ_t and the original admissible smooth weights, the same finite compensated probe satisfies

$$|I_{\eta,\mathrm{modified},t}(Z)| \ll Z^{3/16 - t/4 + \epsilon} \qquad (\epsilon > 0).$$

The constant and threshold may depend on that slot system; the exponent has no mesh loss. No prior zero-free theorem is needed for this implication.

Proof. For a rescaled subset of length d, put $M' = M_t - 2d$ and $\ell' = \ell_t - d$. The retained dual-range calculation keeps $M' + \ell' - 1 = -3d$. The general energy formula gives

$$\sum_{0 < q_m \ll Q} |B_m^J|^2 \ll Z^{M' + s_t(d) + \epsilon}, \qquad s_t(d) = \left(\frac{-1/6 + 5t + d}{4}\right)_+.$$

The extended edition verifies the support mapping and a uniform exponent bound including finite logarithmic offsets; the general arithmetic energy estimate remains a hypothesis. The tuple ratio $r_J = q_{p_J}/Z^d$ stays in a fixed compact subset of $(0,\infty)$, and

$$X' = Z^{l_x-d}/r_J, \quad Y' = Z^{l_y-d}/r_J, \quad Q = q_{b_*}X'Y', \quad P_a = Y'^2/Q = q_{b_*}^{-1}Z^{1/8}.$$

Uniformly over subsets, $l_x - d \ge 11/80$, $l_y - d \ge 21/80$, $M' \ge 2/5$ and $l_y - d - 11/48 \ge 1/30$. Thus $Q, Y', P_a \ge 1$ eventually and $P_a^2/Y' \ll P_a^{1/6}$. The assumed Gram bound and Cauchy–Schwarz give the unscaled tuple bound $(X')^{1/2}P_a^{1/12}Z^{s_t(d)/2+\epsilon}$, with a fixed polynomial height cost integrated against the external smooth Mellin transform. The number of tuples, their coefficient and $X'^{1/2}$ contribute $d - 3d/2 - d/2 = -d$. Since $t \le 1/30$ and $d \ge 0$,

$$s_t(d) \le d/4, \qquad -d + s_t(d)/2 \le -7d/8 \le 0.$$

Sum the finitely many subsets after allocating the small losses. The final exponent is $l_x/2+(1/8)/12 = 3/16 - t/4$. □

The potential row loss must be retained in this proof, rather than set to zero for every subset. The real-power absorption and reflected exponent inequalities are kernel-checked components; the arithmetic energy and Gram bounds are not. This closes the conditional low interface at $t = 10^{-5}$, leaving the nonprincipal contours and the deep source inputs unverified. It gives no unconditional new zero-free theorem.

13 The nonprincipal central-contour interface

The principal second Euler region does not govern the following move. Write $\mathcal{D}_1(\varepsilon_0)$ for the region $\operatorname{Re} s \ge 51/100$, $\operatorname{Re} z \ge 17/50$, $\operatorname{Re} w \ge -1/100$, $\operatorname{Re}(s+w) \ge 1+\varepsilon_0$.

Proposition 13.1 (Conditional central accounting)**.** *Retain the exact arithmetic high identity, full-correction holomorphy and polynomial height bounds on $\mathcal{D}_1$, the source's buffered detector bounds, and both integrated and trace-tail contracts. Fix $\beta_* > \sigma_0 \ge 51/100$ and a retained nonprincipal bin of physical rows $q_u \asymp U = Z^d$, with fixed $0 < d_{\min} \le d \le d_{\max} < \infty$ and $51/100 \le a \le \beta_* \le 1$. Exclude the detector's bounded exceptional rows. Take $T_1 = Z^\tau > 2$, with positive τ within $d_{\min}/100$ and the retained detector height ceiling. For $0 < e < 10^{-3}$ the original bin integral moves to*

$$\operatorname{Re} s = a + 16e, \qquad \operatorname{Re} w = 1 - a - 6e, \qquad \operatorname{Re} z = z_0 = 17/50$$

with error $O_{\eta,N}(Z^B T_1^{-N})$. Here B and internal height orders are fixed before N. On these retained contours suppose an admissible finite decomposition and pointwise row partitions satisfy

$$\sum_{u\in\mathcal{C}} |L^S(w,\chi(u))\mathfrak{H}_u^{(\lambda)}| \ll U^{R+\delta/2+\epsilon_c} Z^{\ell(z_0-1/2)+g+\epsilon_c}(1+T_1)^A,$$

uniformly, with $\epsilon_c \ge 0$, bounded R, g and polynomial logarithmic multiplicity. For every $\epsilon_d, \epsilon_p > 0$, the original bin contribution is bounded by

$$Z^{C(\sigma_0)+\sup E_{\sigma_0}(d;R,g)+(16-6l_y)e+(1+d)\epsilon_c+d\epsilon_d+\epsilon_p}(1+T_1)^A + O_{\eta,N}(Z^B T_1^{-N}),$$

where $\delta = 2a-1$, $h = 1 - l_x + \ell$ and

$$E_\sigma = a - \sigma + h(z_0 - 1/6) - al_y - \ell/2 + g + d(R + \delta/2 - z_0).$$

Auxiliary retained coordinates and discarded integrations must obey the same single cumulative buffered-height allocation.

Proof. Move the full holomorphic tuple correction before introducing pieces. First move z to z_0 and s to $\beta_* + 20e$ with $w = 3$. The reciprocal stays to the right of the actual supremum β_*. Next move w to $1 - a - 6e$; since $a \le \beta_*$, $\mathrm{Re}(s+w) \ge 1 + 14e$. Discard the original height tails while the reciprocal is global, then move the retained s segment to $a + 16e$. The rectangle has $\mathrm{Re}(s + w) \ge 1 + 10e$ and stays in $\mathcal{D}_1$. The buffered hypotheses exclude denominator zeros there. The numerator is entire and nonprincipal, and $6z_0 > 1$, so no pole is crossed. Horizontal joins use the trace contract; the buffered s-axis is never extended beyond its buffer.

Only after this move apply triangle and the aggregate bounds at each point. Dominate all occurring powers by one supremum and integrate the original analytic row sum; no individual pointwise partition is integrated. The outside power is $l_x(1/2 - z_0) + a + z_0 - 1 - al_y - dz_0 + (16 - 6l_y)e$. Adding the assumed row power and subtracting $C(\sigma_0) = \sigma_0 + l_x/2 - 1 + h/6$ proves the formula. □

On the retained region only, local nonvanishing permits the main/error factorization with $g = q\ell$. Strict ramified errors spend the conductor deficit simultaneously: for distinct labels of valuation $j_p \ge 2$, $A_* = 1/2 - w_r \ge 0$ gives $-w_r - (j_p - 1)A_* \le -1/2$. The extended edition records the allocation. For the perturbed geometry, at fixed algebraic δ, q, R, d,

$$E_t(d) - E_0(d) = t(13/50 + \delta/4 + q).$$

Thus the conditional central interface covers $t = 10^{-5}$ without a new 7/8 restriction. It establishes neither negativity of every row exponent nor the underlying arithmetic inputs, outer rows or GRH.

14 A finite arithmetic input: local correlations

The following classical finite-field calculation is unconditional and is verified for actual finite fields and multiplicative characters. It does not establish a global character moment or GRH.

Theorem 14.1 (Zero-preserving local kernel)**.** *Let k be a finite field of size P, and let $A : k \to \mathbb{C}$ be a multiplicative character extended by zero at 0, including when principal. Put*

$$K_A(a, b; t) = \sum_{ax - by = t} A(x)\overline{A(y)}.$$

For $a, b \ne 0$,

$$K_A(a, b; t) = A(b/a) \begin{cases} P - 1, & t = 0, \\ P - 2, & t \ne 0,\ A = 1, \\ -1, & t \ne 0,\ A \ne 1. \end{cases}$$

If exactly one coefficient is zero, the value is $(P-1)1_{A=1}1_{t\ne 0}$. If both vanish, the value is $(P-1)^2$ when $A = 1, t = 0$, and zero otherwise.

Proof. The autocorrelation $C_A(h) = \sum_y A(y+h)\overline{A(y)}$ equals $P - 1$ at $h = 0$. For $h \ne 0$, scale $y = hz$ to reduce to $h = 1$. Changing z to $-z$ gives $C_A(1) = A^{-1}(-1)J(A^{-1}, A)$. The Jacobi identities give -1 for nonprincipal A and $P - 2$ for principal A; the pinned formal proofs are recorded in [13]. For $a, b \ne 0$, solve $x = (t + by)/a$ and factor $A(b/a)$ to obtain $A(b/a)C_A(t/b)$. If $b = 0, a \ne 0$, the fixed value $x = t/a$ and the remaining character sum give the stated result. Swapping x, y and conjugating gives the other case. With both coefficients zero, the sum is $|\sum A|^2$ at $t = 0$ and empty otherwise. □

For a good prime $\mathfrak{p}$ and common exponent $c \geq 1$, take $A = \chi_{\mathfrak{p}}^c$ and $(a, b) = (n_2, n_1)$ in the residue field. On the genuine coprime-residual locus at least one coefficient is a unit. Each field solution has exactly P^{c-1} lifts modulo $\mathfrak{p}^c$, so the common-prime local factor is $P^{c-1}K_A(a, b; t)$. Its norm is at most P^c. A residual modulus meeting the common prime introduces no extra multiplicity. CRT multiplies these local congruence sums.

The restriction matters. For the principal character on $\mathbb{F}_7$, $K_1(0, 0; 0) = 36 > 7$, an actual sum checked in Lean. Thus the genuine kernel bound cannot hold on all residual pairs. The source explicitly uses an artificial zero extension off the coprime-residual locus; this is not a refutation of its stated finite identity.

The Gauss normalization is also consistent. For nonprincipal A and nontrivial additive ψ, the checked finite identity is $|g(A, \psi)|^2 = P$. Choose the fixed primary generator π of $\mathfrak{p}$. Splitting a residue as $x = x_0 + \pi y$ gives the factor $P^{n/2-1}$ after additive orthogonality. Therefore

$$|G(\mathfrak{p}^n, t)| = P^{(n-1)/2} 1_{v_{\mathfrak{p}}(t)=n-1} \quad (6 \nmid n),$$

For $6 \mid n$ the value, including its sign, is $G(\mathfrak{p}^n, t) = P^{n/2} 1_{\mathfrak{p}^n|t} - P^{n/2-1} 1_{\mathfrak{p}^{n-1}|t}$. Finite-field Gauss and kernel identities are formalized; quotient-ring lifting, CRT and the source's arithmetic phase identification remain prose. No full Gram, reflection or moment estimate follows from this finite input alone.

15 Periodic cancellation in the Gram estimate

Lemma 15.1 (Annular periodic Poisson)**.** *Let $\mathcal{O} = \mathbb{Z}[\omega] \subset \mathbb{C}$, $r \in \mathcal{O} \setminus \{0\}$, $q = |r|^2$, and let $a : \mathcal{O}^2 \to \mathbb{C}$ be periodic modulo r in both columns, with $|a| \leq M$ and complete mean zero. Let $W \in C_c^\infty(\mathbb{C}^2)$ be supported on fixed annuli $0 < c \leq |z_j| \leq C$, with sufficiently many fixed seminorms bounded by K. For $N > 0$ and $A > 2$,*

$$\left| \sum_{m \in \mathcal{O}^2} a(m) W(m/\sqrt{N}) \right| \ll_{A,c,C} KMN^2 \min\{1, (q/N)^A\}.$$

Proof. Normalize the finite Fourier coefficient by q^{-2}, the reciprocal number of residue pairs. Its norm is at most M, and its trivial coefficient is zero. Fourier inversion followed by lattice Poisson therefore gives N^2/v^2 times one sum over the dual of $(r\mathcal{O})^2$, where v is the covolume of $\mathcal{O}$. There is no additional q^2 factor. For $N \geq q$, remove the zero frequency and use Fourier decay of order $2A > 4$. The dual spacing is comparable to $q^{-1/2}$, uniformly in the angle of r, giving $\sum_{\xi \neq 0} (1 + \sqrt{N}|\xi|)^{-2A} \ll (q/N)^A$. For $N < q$, fixed annular support makes the original sum empty below a fixed positive N, and otherwise allows $O(N^2)$ lattice pairs. □

The hypothesis at small N matters: a mean-zero periodic coefficient with $a(0) = 1$ and a smooth bump with $W(0) = 1$ can leave the single origin contribution as $N \to 0$. The source weights are annular; this example refutes dropping that condition, not the source estimate.

Proposition 15.2 (Conditional Gram reconstruction)**.** *Retain the September source's exact column/row Poisson representation, artificial common kernel, original zero masks and fixed reciprocity/ray period identification. Assume the stated joint smooth seminorm bounds. Retain its prescribed fixed polynomial ranges, in particular $Z \geq 2$ and $1 \leq Q, Y' \leq Z^{K_0}$ for fixed K_0. For its fixed ray coefficients, put $P = Y'^2/Q \geq 1$. Then the periodic cancellation and positive dyadic summation give*

$$\sum_{q_m \ll Q} |A_m|^2 \ll (1 + |\nu|)^J \frac{Q}{Y'} \left(1 + P^{1/6} + \frac{P^2}{Y'}\right) Z^\epsilon.$$

Proof. Insert the full Möbius sum before separating residual columns. For a nonexceptional frequency k, CRT isolates a nonprincipal residue-field character; its sum is zero. The joint period satisfies $q_r \ll q_C q_k$. In the shell $q_C q_k \asymp RP$, with dyads $q_C \asymp C_0$, $q_d \asymp D_0$, use $N = Y'/(C_0 D_0)$ in Lemma 15.1. The label counts and outer normalization give

$$\frac{Q}{Y'^3}\left(C_0 D_0 \frac{RP}{C_0}\right) C_0 \left(\frac{Y'}{C_0 D_0}\right)^2 = \frac{Q}{Y'}\frac{RP}{C_0 D_0}.$$

For $\Lambda = Y'/(RP)$ and $A \geq 2$, the remaining dyadic sum is

$$S_A(\Lambda) = \sum_{n\geq 0}(n+1)2^{-n}\min\{1, (2^n/\Lambda)^A\} \ll \frac{1+\log(2+\Lambda)}{\Lambda}.$$

Indeed, if $2^L \leq \Lambda < 2^{L+1}$, the head is at most $[L2^{L+1}+1]/\Lambda^2$ and the tail equals $(L+3)2^{-L}$; if $\Lambda < 1$, the sum is 4. Summation in R with a sufficient smooth decay gives P^2/Y'. Exceptional k have sixth-power ideals apart from primes of C and a fixed modulus; their count is $\ll_\epsilon (RP/C_0)^{1/6} q_C^\epsilon$. The trivial column bound leaves summable factors $C_0^{-1/6+\epsilon} D_0^{-1}$, giving $P^{1/6}$. The exact zero-frequency diagonal gives 1. □

Nine Lean declarations check the finite normalized Fourier bound, zero mode, independent-coordinate cancellation, exact dyadic head, infinite total and tail, head inequality and scale identity. The lattice argument above is prose. Applying it to the source still requires the listed arithmetic representation and fixed period; neither the full upstream theorem nor a GRH scope is certified by this reduction.

16 A concrete fixed arithmetic period

The arithmetic period retained in Proposition 15.2 can be supplied in prose using classical reciprocity and class field theory. This does not certify its other analytic representation hypotheses.

Proposition 16.1 (A common defining modulus)**.** *Let S be the fixed source prime set, including primes above 6, and let $\mathfrak{m}_{\mathrm{pres}}$ contain the full defining moduli of its fixed ray/coefficient presentations, including redundant zero primes. Put*

$$\mathfrak{m}_0 = (6)^6 \prod_{\mathfrak{p}\in S} \mathfrak{p}^{25}, \qquad \mathfrak{l} = \mathrm{lcm}((12), \mathfrak{m}_{\mathrm{pres}}, \mathfrak{m}_0).$$

The source's artificial two-column Gram coefficient at fixed $C, d, k \neq 0$ is periodic in each column modulo

$$\mathfrak{r} = \mathrm{lcm}(\mathfrak{l}, \mathrm{rad}(C), \mathrm{rad}(k)), \qquad N\mathfrak{r} \leq N\mathfrak{l}\, q_C q_k.$$

If some prime $p \nmid C\mathfrak{l}$ has $6 \nmid v_p(k)$, its complete joint mean is zero. The defining modulus $\mathfrak{l}$ is independent of C, d, k and Z.

Proof. For $a = u\prod_{\mathfrak{p}\in S}\pi_{\mathfrak{p}}^{e_{\mathfrak{p}}}$, $0 \leq e_{\mathfrak{p}} \leq 5$, the extension $E = F(a^{1/6})/F$ is cyclic of degree dividing six. Its power-residue character has conductor dividing the relative discriminant by the classical conductor–discriminant formula [14, Theorem V.3.27]. The minimal polynomial g is a monic integral factor of $X^6 - a$. The field discriminant divides the power-basis discriminant, and $\mathrm{disc}(gh) = \mathrm{disc}(g)\,\mathrm{disc}(h)\,\mathrm{Res}(g,h)^2$. Thus its conductor divides $(6)^6(a)^5$, hence $\mathfrak{m}_0$. This also covers reducible $X^6 - a$. Arithmetic Frobenius identifies this character with the sextic residue symbol [14, Sections VIII.5.3–5.5]. Actual S-valuations reduce modulo six on S-units.

For odd $c = a + b\omega$, Gaussian Poisson summation gives

$$\Gamma(c) = \frac{1 + i^{-b} + i^{a} + i^{b-a}}{2}.$$

Its values on square classes $1, -1, \lambda, -\lambda$ modulo 4 are $1, 1, i, -i$. The ratio $\Gamma(ab)/(\Gamma(a)\Gamma(b))$ is the symmetric sign bicharacter $\mathcal{R}(a, b)$. Classical cubic reciprocity [16, Equation (1.4)] and CRT for quadratic Gauss sums identify it with $\chi_b(a)/\chi_a(b)$ on coprime primary pairs. For general pairs its separate finite definition is retained.

Write $k = \zeta k_S k_{\mathrm{good}}$. Reciprocity gives, with every zero retained,

$$\chi_m(k) = \kappa_{\zeta,\boldsymbol{e}}(m)\mathcal{R}(k_{\mathrm{good}}, m) \prod_{p \mid k_{\mathrm{good}}} \chi_p(m)^{e_p}.$$

The first factor has conductor dividing $\mathfrak{m}_0$, the second has period 4, and the product has period $\mathrm{rad}(k_{\mathrm{good}})$. Fixed coefficients use $\mathfrak{l}$ and the artificial kernel uses $\mathrm{rad}(C)$. The substitutions by C, d introduce no larger period. All nonunit/primary zero branches are taken before inverses. At a nonexceptional prime outside $C\mathfrak{l}$, one nonprincipal field coordinate remains independent of all other factors; its mean is zero. □

The larger bad-prime powers are essential. The primary primes $\pi_{31} = 1 + 6\omega$ and $\pi_{43} = 1 - 6\omega$ agree modulo 12, but

$$\chi_{\pi_{31}}(\lambda) = \zeta_6, \qquad \chi_{\pi_{43}}(\lambda) = \zeta_6^5, \qquad \zeta_6 = 1 + \omega.$$

Indeed $\omega = 5$ in $\mathbb{F}_{31}$ and $\omega = 36$ in $\mathbb{F}_{43}$, giving $11^5 = 6$ and $30^7 = 7$ in the respective fields. Thus modulus 12 for the quadratic phase cannot replace the full fixed numerator modulus. The source already allows those higher powers. Thirteen Lean components check the actual quotient-algebra phase and period; the Gaussian, cubic-reciprocity and conductor arguments remain classical prose dependencies, with no zero-location assumption.

17 Arithmetic extraction in a reflected block

Proposition 17.1 (Marked coefficient separation)**.** *Assume the canonical completed-theta reflection formula with its stated cusp support and coefficient bound. Retain the source block hypotheses: disjoint slot supports, original product-form slot coefficients independent of rows and dual variables, and fixed row restrictions independent of the active tuples and dual variables except the explicit (P, R) mask. Freeze every nonresidual base prime, including every $j = 4$ prime, and fix separate full M_{ref}^2 sectors of the residual row product R and active marked product P. Then the reflected block has the coefficient independences of the explicit quadratic–cubic norm in the source's equation (4.6a), with every original zero retained. Its per-tuple central exponent is*

$$-\frac{v}{2} - \ell_b - \frac{e_\lambda}{3} - \frac{S_0 + B_0 + z_a}{2}.$$

This is a conditional arithmetic reduction, not the hybrid mean-square bound.

Proof. At an active prime write $t = c/p$. It is a unit, and $\sigma_p = \lambda^2 t$, $\epsilon_p = -\lambda^{-5} t^{-2}$. The local scalar factors as $\xi_{p,j}\chi_p(c/p)^{2j+2}$, with $\xi_{p,j}$ depending on that prime and the fixed datum only. This follows

directly from the two unit powers; at $j = 4$ the exponent is $-2 \equiv 2j + 2 \pmod 6$. For distinct active primes, cubic reciprocity gives

$$\chi_p(q)^{2j+2}\chi_q(p)^{2k+2} = \left(\frac{q}{p}\right)_3^{j+k+2}.$$

Thus residual $j = 1$ and marked $k = 0$ phases cancel. All other prime interactions and the angular conductor factor separate into row, tuple or fixed scalars. The full sectors fix the cusp data.

The moving columns, including their zeros, are

$$\chi_R(nb^3)^3 = \chi_R(nb)^3, \qquad \chi_P(nb^3)^{-2} = \chi_P(n)^{-2}1_{(P,b)=1}.$$

A frozen Ramanujan divisibility term is partitioned into $p \mid n_0$ and $p \nmid n_0,\ p \mid b_0$. In the first case extract only its squarefree occurrence; in the second extract one cube occurrence. For $\mu = u\lambda^h N_F n(B_F b)^3$, the coefficient bound gives

$$\frac{d(\mu)\alpha(\mu)\vartheta(\lambda^4\mu)}{\sqrt{q_\mu}} = 27\delta(n,b)q_{\lambda^h}^{-1/3}q_{N_F n}^{-1/2}q_{B_F b}^{-1}, \quad |\delta| \le 1.$$

The extracted Ramanujan roots leave precisely $(S_0 + B_0)/2$; marked roots give $z_a/2$. The scalar $27/81$ is $1/3$. Shared cube factors remain in b and δ. Consequently the dual coefficient is independent of R, P, while the bounded tuple coefficient is independent of R, n, b. Allowed source primes of S in n, b are retained. □

The freezing condition is necessary for this separation. With primary $p = 1 + 3\omega$, $q = 4 + 3\omega$, the sextic phases are $a = \chi_p(q) = \zeta_6$, $b = \chi_q(p) = \zeta_6^4$. Then $a^4b^2 = 1$, whereas $a^4b^{10} = \omega \neq 1$ for a moving $j = 4$ prime. The source already freezes it. Likewise deleting $1_{(P,b)=1}$ changes the local value at $n = 1, b = 0$ from zero to one.

For completeness, the common-profile saving has a separate analytic proof. Let V and all its Euler derivatives decrease faster than every real power at zero and infinity. Put $\widehat{V}(t) = \int_0^\infty V(x)x^t\,dx/x$, $K = (2\pi)^4/27$, and

$$R(t) = \prod_{\pm}\frac{\Gamma(1 + t \pm 1/6)}{\Gamma(1 - t \pm 1/6)}, \qquad V^\sharp(x) = \frac{1}{2\pi i}\int_{(0)}\widehat{V}(-t)R(t)(Kx)^{-t}\,dt.$$

The finite seminorm used below is

$$\mathfrak{M}_{A,B}(V) = \sup_{-A\le\eta\le 1/4}\int_{\mathbb{R}}(1 + |u|)^B|\widehat{V}(\eta + iu)|\,du.$$

The gamma quotient is holomorphic for $\Re t > -5/6$ and satisfies $|R(\sigma + iu)| \ll_A (1 + |u|)^{4\sigma}$ on $-1/4 \le \sigma \le A$ by [17]. Rapid Mellin decay permits shifts to $-1/4, 0, A$, giving

$$|D_x^j V^\sharp(x)| \ll_{A,j} \min\{x^{1/4}, (1 + x)^{-A}\}\mathfrak{M}_{A,\lceil 4A\rceil + j + 2}(V).$$

On a fixed logarithmic box, $F_Y(\boldsymbol{y}) = W(\boldsymbol{y})V^\sharp(Y\prod y_i^{a_i})$ has one Fourier density common to every row and tuple. Euler derivatives introduce fixed a_i factors, with no extra Y power. Extract $m_{\mathrm{ker}} = Z^{-(T_d - v - 3\ell_b - e_\lambda)_+/4}$ from that linear vector before Minkowski; its square enters only afterward. Enlarge the positive row set after separation, with no kernel reevaluation on added rows. Nine new Lean components check the local characters, phase algebra and the integrable common-density norm step. The global theta/cusp identification and the hybrid analytic estimate remain separate, unverified inputs; no zero-free or GRH theorem follows.

18 Consequences for a revised research article

Theorems 3.1, 3.2, and 3.3 require replacements for the associated proof steps, not additional exposition. A credible next theorem would construct operators independently of unknown zeros and establish their exact arithmetic determinant, including ramified, archimedean, and conductor terms. It must then prove locally uniform convergence or the local boundedness in Proposition 2.3. Only then does Theorem 2.1 prove the desired zero location.

Weil/Li positivity, adelic trace methods, and canonical-product approximations are potential alternative interfaces. They move the research obligation to a specific positivity, trace, or approximation theorem; they do not discharge it by analogy. Lagarias [4] and Connes [5] provide relevant primary-source starting points. No novelty or complete proof is claimed for these classical conditional principles.

19 Why an absolute coprime pair-square estimate cannot close the short rows

Keep the original sextic family

$$M_u(D) = D^{-1/2} \sum_n \mu(n)\nu(n)W(Nn/D)\chi_n(u), \qquad \chi_n(u) = (u/n)_6,$$

with squarefree primary column ideals outside the fixed excluded set, their full annular coefficients (fix W supported in $[1,2]$), and all nonzero Eisenstein element rows $Nu \le H$. The cubic symbol is χ_n^2, not χ_n. The following classical frame inequality [18] gives a new obstruction to one proposed estimate in this arithmetic program.

Proposition 19.1 (Coprime pair-square obstruction)**.** *Let A be an R-by-N complex matrix with $|A_{ui}| = 1$, and put $K_{ij} = \sum_u A_{ui}\overline{A_{uj}}$. Then*

$$\sum_{i \ne j} |K_{ij}|^2 \ge NR(N-R).$$

For the original character kernels, let $B_1(H,D)$ be the sum of $|K_H(n,m)|^2$ over all distinct coprime squarefree column ideals in $[D,2D]$. At $H = D^{3/4}$,

$$B_1(H,D) \gg \frac{D^{11/4}}{(\log D)^2}.$$

In particular the proposed sufficient input $B_1(H,D) \ll H^2 D^{\varepsilon}$ fails for each fixed $\varepsilon < 5/4$. This does not refute the signed Möbius mean square.

Proof. Set $F_{uv} = \sum_i A_{ui}\overline{A_{vi}}$. Reordering four finite sums proves $\sum_{i,j} |K_{ij}|^2 = \sum_{u,v} |F_{uv}|^2$. The row diagonal contributes RN^2; the column diagonal contributes NR^2. Removing the latter and retaining the former gives the inequality.

Restrict the column pairs in the nonnegative sum B_1 to distinct prime ideals p, q of norm in $[D,2D]$, outside the same fixed set. If $H < D$, no such prime divides a nonzero row element: divisibility would imply $Nu \ge Np$. Thus all original zero masks are exactly one on this ball, $|\chi_p(u)| = 1$, and the residual primitive finite character conductor is pq, of norm between D^2 and $4D^2$. All these pairs have common divisor1. The lattice count gives $R = (2\pi/\sqrt{3})H + O(\sqrt{H}+1)$. The classical unconditional fixed-field prime-ideal theorem gives $N \sim D/\log D$; this is an external analytic input, not a GRH hypothesis. At the stated endpoint $R = o(N)$, and the bound is therefore $(1-o(1))N^2R \gg D^{11/4}/(\log D)^2$. □

To see the exact failed step, write $c_n = \mu(n)\nu(n)W(Nn/D)$ and $C_D = \sum_n |c_n|^2 = O(D)$. Cauchy over coprime pairs bounds the absolute signed contribution by $D^{-1}C_D B_1^{1/2}$. The hoped-for pair-square upper bound would have closed this particular route, but is false. A large absolute pair-square mass is not a lower bound on the quadratic form with the fixed c_n. Nor is the prime part a positive summand of the full Möbius energy: prime–composite cross terms can cancel. The original low-gcd estimate and the positive arithmetic determinant bridge remain open.

20 A signed Möbius decomposition and its termwise obstruction

Keep the original squarefree good-ideal family, its fixed ν, its full $W(Nn/D)$ and every zero-extended sextic symbol. In ideal Dirichlet convolution, write $\zeta_0(n) = 1$ for every good ideal and let δ be the convolution unit. For $1 \le U < D$ put $a_U = \mu \mathbf{1}_{Nn \le U}$ and $b_U = \mu - a_U$. The classical Vaughan algebra gives

$$\mu = 2a_U - a_U * a_U * \zeta_0 + b_U * b_U * \zeta_0. \tag{1}$$

Indeed $\mu * \zeta_0 = \delta$, and expansion of the last product proves the identity without an analytic assumption. This is a classical decomposition, not a new cancellation theorem; compare [19, Lemma4.1].

On squarefree n all factors in $n = dc = abc$ are pairwise coprime. Define the actual, nonarbitrary coefficient functions

$$\beta_U(d) = \sum_{\substack{ab=d \\ Na, Nb \le U}} \mu(a)\mu(b), \qquad g_U(s) = \sum_{\substack{b \mid s \\ Nb > U}} \mu(b).$$

For squarefree d, $\beta_U(d)$ is $\mu(d)$ times the number of the displayed ordered factorizations. For $Ns > U$, $g_U(s) = -\sum_{b \mid s,\, Nb \le U} \mu(b)$. The two regrouped pieces are

$$I_u = D^{-1/2} \sum_{\substack{d,c \text{ squarefree} \\ (d,c)=1}} \beta_U(d)\nu(dc)\chi_d(u)\chi_c(u)W(NdNc/D),$$

$$II_u = D^{-1/2} \sum_{\substack{a,s \text{ squarefree} \\ Na, Ns > U,\, (a,s)=1}} \mu(a) g_U(s)\nu(as)\chi_a(u)\chi_s(u)W(NaNs/D).$$

Then $M_u = -I_u + II_u$. All ideals here remain outside the same fixed excluded set. The short $2a_U$ term vanishes because $U < D$ and W is supported in $[1, 2]$. No phase or coprimality mask was removed.

Proposition 20.1 (An all-cutoff separate-piece bound is false)**.** *Take the allowed principal ν, $U = 1$, and a fixed nonnegative smooth W supported in $[1, 2]$, bounded below by a positive constant on a closed subinterval of $(1, 2)$ of positive length. For each of the two pieces,*

$$\sum_{0 < Nu \le H} |I_u|^2 \gg_W \frac{D}{\log^2 D}, \qquad \sum_{0 < Nu \le H} |II_u|^2 \gg_W \frac{D}{\log^2 D} \quad (H \ge 1).$$

Consequently a proposed bound HD^{ε} for each piece at $H = D^{3/4}$ fails for $\varepsilon < 1/4$. In the exact original notation $H(HD)^{\varepsilon}$, it fails for $\varepsilon < 1/7$. This refutes an all-cutoff termwise strengthening, not the original signed mean square or a specified moving-cutoff argument.

Proof. At $U = 1$, $a_U = \delta$, so $a_U * a_U * \zeta_0 = \zeta_0$ and $b_U * b_U * \zeta_0 = \mu + \zeta_0 - 2\delta$. On the annulus the two coefficients are 1 and $1 + \mu(n)$, respectively. On squarefree ideals the latter is zero or two. At the actual unit row $u = 1$, every original character equals one. A fixed-field prime-ideal theorem

gives $\gg D/\log D$ good prime columns in the subinterval, so $I_1 \gg_W \sqrt{D}/\log D$. Fix a good prime ideal q_0. Products $q_0 p$ with varying distinct good primes in the rescaled subinterval have $\mu(q_0 p) = 1$. Their nonnegative contributions give $II_1 \gg_W \sqrt{D}/\log D$. Keeping that one row in each nonnegative energy proves the result. The prime-ideal theorem is a classical unconditional external input, with the field and q_0 fixed. □

The factorization does not shorten the primitive row conductor: for squarefree coprime a, s it is as, not a or s separately. For two resulting columns n, n' with common factor h, the oscillatory conductor is nn'/h^2 and a zero-preserving period is nn'/h. The signed low-gcd estimate and the independent positive arithmetic determinant comparison therefore remain open.

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